

DSA 210 Final Report: Impact of Brent Oil Prices on Coca-Cola and McDonald's Stock Prices

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1. Motivation

The primary motivation of this project is to investigate the macroeconomic dependencies of global food and beverage giants, specifically Coca-Cola (KO) and McDonald's (MCD). Brent Oil prices act as a major indicator of global economic health, directly influencing transportation, manufacturing, and supply chain costs. By analysing how these operational cost indicators reflect on stock prices, this study aims to uncover the vulnerability or resilience of the consumer goods sector against energy market volatility. Understanding these dynamics provides valuable insights for predictive financial modelling and strategic investment decision-making.

2. Data Source

The data collection process involved merging two primary data streams:

- **Stock Market Data:** Historical daily stock prices for Coca-Cola and McDonald's were collected using public financial APIs (e.g., Yahoo Finance).
- **Macroeconomic Indicator:** Historical Brent Oil prices were sourced and utilized as the primary enrichment dataset (BrentOilPrices.csv).
- **Data Preprocessing:** The datasets were merged on their respective date columns. Missing values (such as non-trading days and market holidays) were handled, and the final structured dataset was saved as `cleaned_data.csv` to ensure consistency across the analysis pipeline.

3. Data Analysis

The analysis pipeline was conducted in Python and split into multiple stages:

- **Exploratory Data Analysis (EDA):** Initial visualizations were created to observe the trends of KO, MCD, and Brent Oil over time. Daily percentage returns (`ko_returns` and `mcd_returns`) were calculated to standardize the stock price movements.
- **Hypothesis Testing (`eda_hypothesis_test.ipynb`):** To statistically validate the relationships, I analysed the daily stock returns. The dataset was split into two regimes based on the median Brent Oil price: "High Oil Price" and "Low Oil Price" periods. I formulated the null hypothesis (H_0) that there is no significant difference in stock returns between these two periods. To test this, I conducted

both Independent Two-Sample T-tests (assuming normality) and Mann-Whitney U tests (as a non-parametric alternative).

- **Machine Learning Models (ML_Methods.ipynb):** To evaluate the predictive power of Brent Oil prices on the daily returns of KO and MCD, I implemented three different regression models: **Linear Regression, Random Forest**
- **Regressor, and XGBoost Regressor.** The models were evaluated using Mean Squared Error (MSE), Mean Absolute Error (MAE), and the R-squared (R^2) metric to determine how much variance in stock returns could be explained by oil prices.

4. Findings

The data analysis, hypothesis testing, and machine learning implementations yielded the following key insights:

- **Statistical Significance:** The hypothesis tests consistently indicated that we fail to reject the null hypothesis. For Coca-Cola, the T-test yielded a T-statistic of -1.2185 with a p-value of 0.2231, and the Mann-Whitney U test yielded a p-value of 0.0968. For McDonald's, the T-test resulted in a p-value of 0.4782, and the Mann-Whitney U test yielded 0.6975.
- **Conclusion on Market Impact:** Since all p-values are well above the standard 0.05 alpha level, there is no statistically significant difference in the daily stock returns of KO and MCD between high and low oil price regimes.
- **Model Performance & Predictive Power:** The machine learning models strongly corroborated the hypothesis test results. Across all models for both KO and MCD, the R^2 scores were essentially zero or slightly negative (e.g., Linear Regression yielded an R^2 of -0.0003 for KO and -0.0005 for MCD; XGBoost yielded -0.0694 for KO and -0.0650 for MCD).
- **Final Verdict:** The negative R^2 values indicate that the models perform no better than simply predicting the mean of the stock returns. This proves that daily Brent Oil price fluctuations contain **no predictive power** for the daily returns of Coca-Cola and McDonald's stocks. While energy costs are vital macroeconomically, daily stock volatility in the food and beverage sector is driven by other overriding factors (e.g., earnings reports, consumer trends, or broader market indices).

5. Limitations and future work

Limitations:

- **Omitted Variable Bias:** Stock prices are influenced by countless factors including inflation rates, corporate earnings reports, and global pandemics. Relying heavily on Brent Oil prices oversimplifies the financial market dynamics.
- **Time Lag:** The models assumed a direct daily impact. In reality, changes in oil prices might take weeks or months to affect corporate balance sheets and stock prices.

Future Work:

- **Feature Enrichment:**

Future iterations of this project should include additional macroeconomic indicators such as the Consumer Price Index (CPI), interest rates, and currency exchange rates.

- **Advanced Modeling:**

Applying time-series specific models like ARIMA, Prophet, or LSTM (Long Short-Term Memory) networks could better capture the sequential dependencies and delayed effects in the financial data.

6. Academic Integrity & AI Usage

During the development of this project, I utilized Large Language Models, specifically Claude and Gemini, to assist with writing Python code, fixing bugs, and verifying the accuracy of the analytical outputs. Furthermore, these AI tools were used during the documentation process to help with page layout, formatting, and general proofreading. Despite this assistance, the core analytical logic, project design, and final implementations are entirely my own original work.